

Numerical and Algebraic Expressions — Distributing and Combining Without Breaking the Rules

Answer Key

Grades 6–7

Quick Answers

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|--------------|--------------|---------------|--------------|
| 1. $4x + 28$ | 3. 19 | 5. 12 | 7. $6x + 15$ |
| 2. 27 | 4. $5a + 13$ | 6. $14x - 15$ | 8. 10 |
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Curriculum

Level: Grades 6–7

6.EE.A / 7.EE.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

- 2. *If they got 17: multiplied only the first term inside the parentheses, leaving the second term alone*
 - 3. *If they got 48: added the coefficients of two unlike terms and multiplied the letters together*
 - 5. *If they got 0: kept the minus sign on the second term instead of distributing the negative two to it*
 - 8. *If they got -10: distributed the first bracket correctly but kept both signs inside the second one*
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1. Write $4(x + 7)$ without parentheses.

Step 1. The 4 outside multiplies everything inside, so there are two products to write, not one: $4 \cdot x$ and $4 \cdot 7$.

Step 2. $4 \cdot x = 4x$ and $4 \cdot 7 = 28$, joined by the same $+$ that was inside. $4x + 28$

Step 3. Check with a number: at $x = 1$, the original is $4(8) = 32$ and the answer is $4 + 28 = 32$ ✓.

2. Kim: $3(x + 5) = 3x + 5$. Test at $x = 4$.

Step 1. Evaluate the original the way the parentheses say: inside first, $4 + 5 = 9$, then $3 \cdot 9 = 27$. 27

Step 2. Evaluate Kim's version at the same x : $3(4) + 5 = 12 + 5 = 17$. The two disagree, so Kim's expression is not equivalent — one disagreement is enough to prove it.

Step 3. Name the error. Kim multiplied only the first term inside the parentheses. Distributing means every term gets multiplied: $3(x + 5) = 3x + 15$, and $3(4) + 15 = 27$ ✓ agrees with Step 1.

Step 4. The gap $27 - 17 = 10$ is exactly the $15 - 5$ that Kim never multiplied — worth pointing out, because it shows the error has a size, not just a name.

3. Devon: $5x + 3y = 8xy$. **Test at** $x = 2, y = 3$.

Step 1. Evaluate the original: $5(2) = 10$ and $3(3) = 9$, so $10 + 9 = 19$. 19

Step 2. Evaluate Devon's version: $8(2)(3) = 48$. Not equal to 19, so the two expressions are different.

Step 3. Name the error. Devon added the coefficients $5 + 3$ AND multiplied the letters. Neither move is legal here: $5x$ and $3y$ are unlike terms because their letter parts differ, so the expression $5x + 3y$ is already as simple as it gets.

Step 4. A model that helps: if x is the price of a pen and y the price of a notebook, $5x + 3y$ is the cost of five pens and three notebooks. There is no single item priced xy , so no single term can hold both.

4. Simplify $7a + 4 - 2a + 9$.

Step 1. Sort the terms into like groups, carrying each sign with the term it belongs to: the a terms are $7a$ and $-2a$; the constants are $+4$ and $+9$.

Step 2. Combine inside each group only: $7a - 2a = 5a$, and $4 + 9 = 13$.

Step 3. $5a + 13$ — and this cannot be shortened further, because $5a$ and 13 are unlike.

Step 4. Check at $a = 2$: the original is $14 + 4 - 4 + 9 = 23$, and the answer is $10 + 13 = 23$ ✓.

5. Nadia: $8 - 2(x - 3)$ **written as** $8 - 2x - 6$; **test at** $x = 1$.

Step 1. Read what is being distributed. It is not 2 but -2 , and it multiplies BOTH terms in the parentheses: $-2 \cdot x = -2x$ and $-2 \cdot (-3) = +6$.

Step 2. So $8 - 2(x - 3) = 8 - 2x + 6 = 14 - 2x$. Nadia kept the second term negative, which is where a sign error of this kind always lives.

Step 3. Evaluate the correct expression at $x = 1$: $14 - 2 = 12$. 12

Step 4. Nadia's version at $x = 1$ gives $2 - 2 = 0$, which is 12 away from the true value — the whole 12 she lost by not distributing the sign.

6. Simplify $5(2x - 3) + 4x$.

Step 1. Distribute the 5 across both terms: $5 \cdot 2x = 10x$ and $5 \cdot (-3) = -15$, giving $10x - 15 + 4x$.

Step 2. Now combine like terms. The x terms are $10x$ and $4x$; the constant -15 has nothing to pair with, so it stays as it is.

Step 3. $10x + 4x = 14x$, so the expression is $14x - 15$

Step 4. Check at $x = 2$: the original is $5(1) + 8 = 13$, and the answer is $28 - 15 = 13$ ✓. Note what must NOT happen in Step 3: $14x$ and -15 are unlike, so they never merge into a single number.

7. Factor $6x + 15$, **then check by distributing back.**

Step 1. Find the greatest common factor of the two terms. $6x = 2 \cdot 3 \cdot x$ and $15 = 3 \cdot 5$, so the largest factor they share is 3.

Step 2. Divide each term by 3 and write what is left inside parentheses: $6x \div 3 = 2x$ and $15 \div 3 = 5$. $3(2x + 5)$

Step 3. Check by distributing back, which is the whole point of the exercise: $3 \cdot 2x = 6x$ and $3 \cdot 5 = 15$, giving $6x + 15$ ✓.

Step 4. This is the same rule read right to left. A student who writes $3(2x + 5) = 6x + 5$ has undone only half of the factoring — and the check in Step 3 catches it immediately, which is why the check is worth doing every time.

8. (a) Why unlike terms cannot combine. (b) Simplify $3(2x - 4) - 2(x - 5)$ and test at $x = 3$.

(a) *Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.*

A letter stands for a number nobody has fixed yet, and x and y may stand for different numbers. Combining $5x$ and $3y$ into one term would force them to count the same thing. The substitution test proves it: at $x = 2$, $y = 3$ the expression is worth 19 while $8xy$ is worth 48, and a single disagreement is enough. Terms may be added only when their letter parts are identical, because then they really are copies of the same unknown amount.

What a full-credit answer must contain: the idea that a letter stands for an unknown number, that unlike terms count different things, and (ideally) one substitution that shows the two expressions disagree.

(b) Step 1. Distribute each bracket, signs included: $3(2x - 4) = 6x - 12$, and $-2(x - 5) = -2x + 10$ — the second one is where the sign error lives, since $-2 \cdot (-5) = +10$.

Step 2. Combine like terms: $6x - 2x = 4x$ and $-12 + 10 = -2$, so the simplified expression is $4x - 2$.

Step 3. Substitute $x = 3$ into the simplified expression: $4(3) - 2 = 10$, and the original gives $3(2) - 2(-2) = 6 + 4 = 10$ ✓. 10

Step 4. The classmate who keeps both signs inside the second bracket writes $6x - 12 - 2x - 10 = 4x - 22$ and gets -10 at $x = 3$ — same first bracket, same combining, one sign.